Chapter 9: Linear Momentum

The following subjects will be discussed

- Center of Mass.
- Linear Momentum.
- Linear Momentum Conservation Law.
- Linear Momentum and Kinetic Energy.
- Collisions.

Linear Momentum and Kinetic Energy

We try to answer the following question.

1) How is the change in momentum related to the change in kinetic energy?
2) For an isolated system, if momentum is conserved is KE conserved as well?

Consider a simple 2-particle collision problem in 1D.

Before collision m_1, v_1^i and m_2, v_2^i After collision m_1, v_1^f and m_2, v_2^f

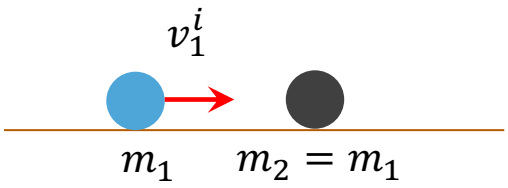
	Elastic collision	Inelastic collision
P	$m_1 v_1^i + m_2 v_2^i = m_1 v_1^f + m_2 v_2^f$	$m_1 v_1^i + m_2 v_2^i = m_1 v_1^f + m_2 v_2^f$
KE	$\frac{1}{2} m_1 v_1^{i2} + \frac{1}{2} m_2 v_2^{i2} = \frac{1}{2} m_1 v_1^{f2} + \frac{1}{2} m_2 v_2^{f2}$	$\frac{1}{2} m_1 v_1^{i2} + \frac{1}{2} m_2 v_2^{i2} = \frac{1}{2} m_1 v_1^{f2} + \frac{1}{2} m_2 v_2^{f2} + loss$

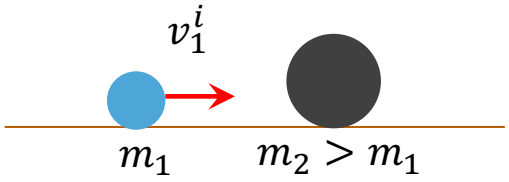
The inelastic case is the general case
In both cases, momentum is conserved

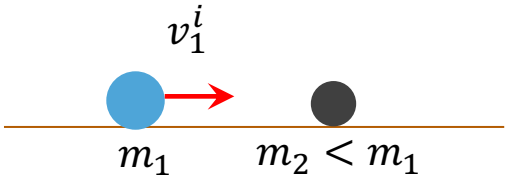
Collision

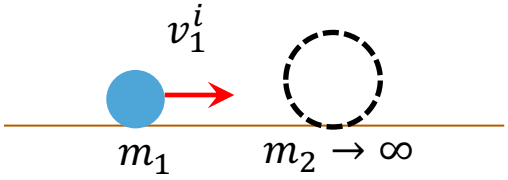
We can study several collision cases.

Stationary Target

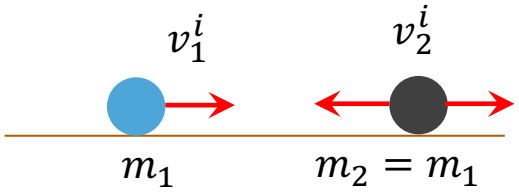


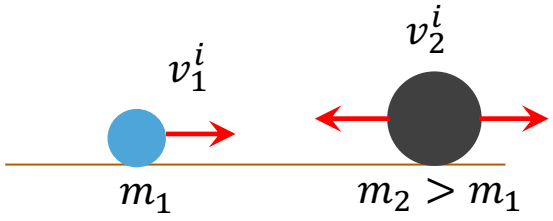


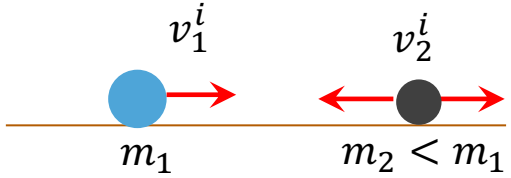


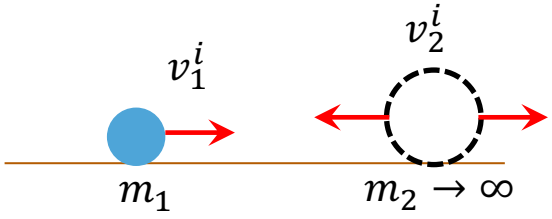


Moving Target

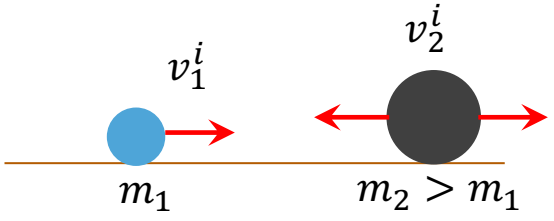


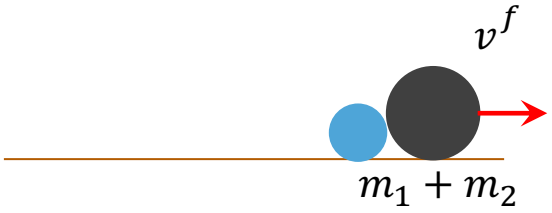






Completely inelastic





Elastic Collision

Starting with the two equations

$$m_1 v_1^i + m_2 v_2^i = m_1 v_1^f + m_2 v_2^f \quad \Rightarrow \quad m_1(v_1^i - v_1^f) = -m_2(v_2^i - v_2^f) \qquad 1$$

$$\frac{1}{2}m_1 v_1^{i2} + \frac{1}{2}m_2 v_2^{i2} = \frac{1}{2}m_1 v_1^{f2} + \frac{1}{2}m_2 v_2^{f2} \quad \Rightarrow \quad m_1(v_1^i - v_1^f)(v_1^i + v_1^f) = -m_2(v_2^i - v_2^f)(v_2^i + v_2^f) \qquad 2$$

Divide 1/2

$$v_1^i + v_1^f = v_2^i + v_2^f$$

Can be written in a more useful form

2-particle elastic collision in 1D

$$v_1^f = \frac{m_1 - m_2}{m_1 + m_2} v_1^i + \frac{2m_2}{m_1 + m_2} v_2^i$$

$$v_2^f = \frac{2m_1}{m_1 + m_2} v_1^i + \frac{m_2 - m_1}{m_1 + m_2} v_2^i$$

Center of mass dynamic (isolated system)

$$v_{cm}^i = \frac{m_1 v_1^i - m_2 v_2^i}{m_1 + m_2} = \frac{m_1 v_1^f - m_2 v_2^f}{m_1 + m_2}$$

Same before and after

Stationary target $v_2^i = 0$

Case	v_1^f	v_2^f
$m_2 = m_1$	0	v_1^i
$m_2 > m_1$	$< \lvert v_1^i \rvert$	$\frac{2}{1 + m_2/m_1} v_1^i$
$m_2 \to \infty$	$v_1^f = -v_1^i$	0
$m_2 < m_1$	$v_1^f < \lvert v_1^i \rvert$	$\frac{2}{1 + m_2/m_1} v_1^i$
$m_2 \to 0$	$v_1^f = v_1^i$	$2v_1^i$

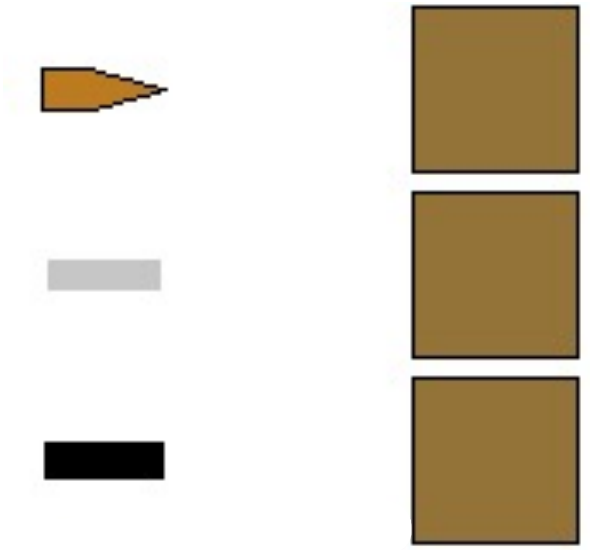
Completely Inelastic Collision

The momentum equation becomes

$$m_1 v_1^i + m_2 v_2^i = (m_1 + m_2) v^f$$

For stationary target

$$v^f = \frac{m_1}{m_1 + m_2} v_1^i$$



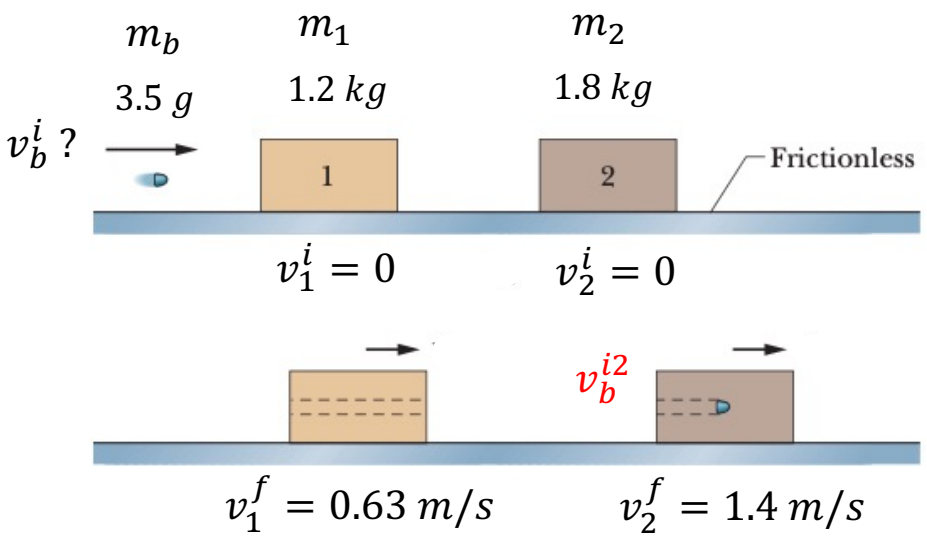
Examples

1) Find the initial bullet velocity (just before hitting object 1).

Complete inelastic collision with object 2.

$$v^f = 1.4 = \frac{m_b}{m_b + m_2} v_b^{i2}$$

$$v_b^{i2} = 1.4 \frac{m_b + m_2}{m_b} = 721.4\ m/s$$



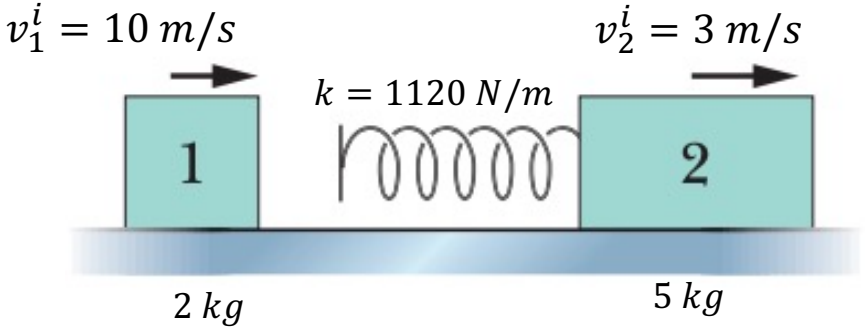
Conservation of momentum for object 1-bullet system.

$$m_1 v_1^i + m_b v_b^i = m_1 v_1^f + m_b v_b^f \quad \Rightarrow \quad m_1(0) + m_b v_b^i = m_1 v_1^f + m_b v_b^{i2}$$

$$(0.0035)\, v_b^i = (1.2)(0.63) + (0.0035)(721.4) \quad \Rightarrow \quad v_b^i = 937.4\ m/s$$

Examples

2) Object 1 causes the spring to have maximum compression, then the two objects move together.
What is the maximum compression?



Complete inelastic collision with object 2.

$$m_1 v_1^i + m_2 v_2^i = (m_1 + m_2) v^f$$

$$v^f = \frac{m_1 v_1^i + m_2 v_2^i}{m_1 + m_2} = \frac{2(10) + 5(3)}{2 + 5} = 5\ m/s$$

Potential energy stored in spring = change in kinetic energy

$$\frac{1}{2} k x^2 = \Delta KE = \left(\frac{1}{2} m_1 v_1^{i2} + \frac{1}{2} m_2 v_2^{i2}\right) - \frac{1}{2}(m_1 + m_2) v^{f2} = 122.5 - 87.5 = 35\ J$$

$$x = 0.25\ m$$

Collisions in Two Dimensions

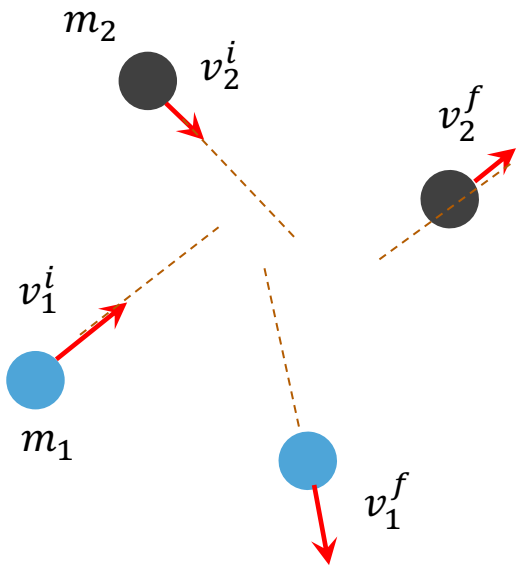
For a closed and isolated system, linear momentum is conserved.

$$m_1\vec{v}_1^{\,i} + m_2\vec{v}_2^{\,i} = m_1\vec{v}_1^{\,f} + m_2\vec{v}_2^{\,f}$$

The above equation can be split into two independent equations one for each direction.

If the collision is elastic, KE is conserved.

$$KE_1^i + KE_2^i = KE_1^f + KE_2^f$$



Examples

1) Find the velocities after collision.

$$m\,\vec{v}_1^{\,i} + m\,\vec{v}_2^{\,i} = m\,\vec{v}_1^{\,f} + m\,\vec{v}_2^{\,f}$$

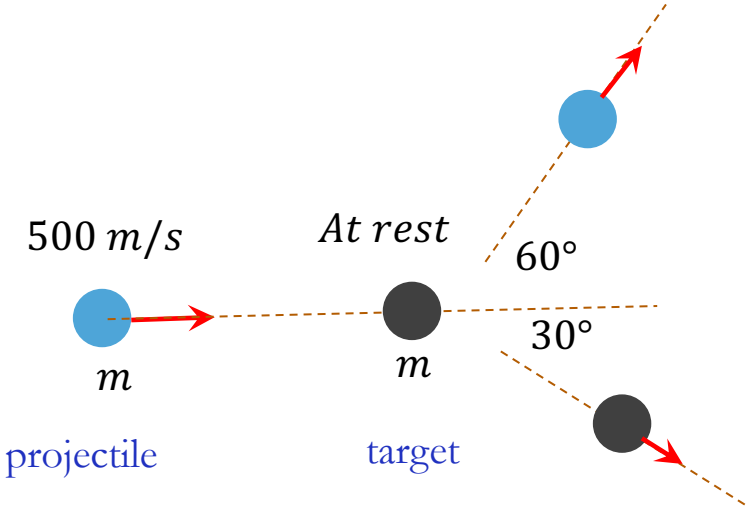
$$\vec{v}_1^{\,i} + (0) = \vec{v}_1^{\,f} + \vec{v}_2^{\,f}$$

$$v_{1x}^{i} = v_{1x}^{f} + v_{2x}^{f}$$

$$500 = v_1^f \cos 60 + v_2^f \cos 30 \quad (1)$$

$$v_{1y}^{i} = v_{1y}^{f} + v_{2y}^{f}$$

$$0 = v_1^f \sin 60 + v_2^f \sin 30 \quad (2)$$



Solve 1 and 2 simultaneously. $v_1^f = 250\ m/s$ $v_2^f = 433\ m/s$

Chapter 10: Rotational Motion

The following subjects will be discussed

- Rotational/angular Motion Variables.
- Equations of Motion.
- Inertia of an Object.
- Kinetic Energy.
- Torque and Newton's Laws.
- Work in Angular Motion.

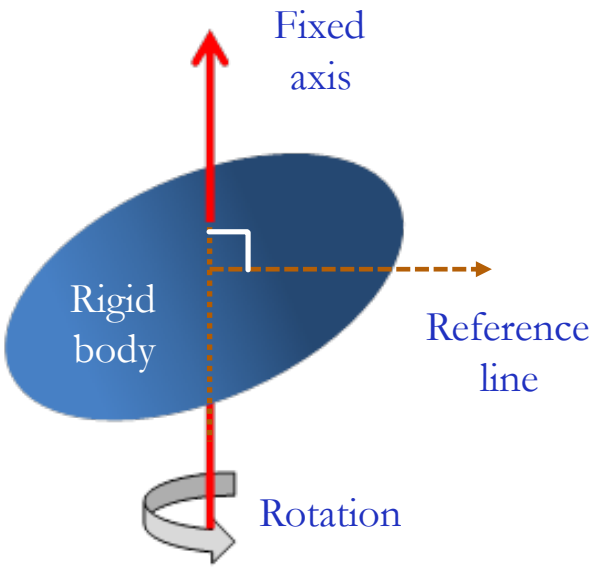
Angular Motion Variables

Consider a rigid body rotating around a fixed axis. How do describe its motion (distance, displacement, velocity, acceleration, etc.)?

Object rotates all together (all parts)

Axis is fixed (in-place rotation)

Each point on object makes a circular trace.



1) Position- Distance- Displacement

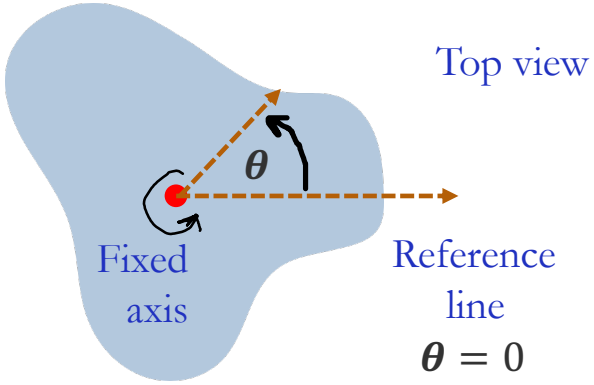
We take $\theta(t)$ as the position at any time. Measured in radians. 1 rad = 0.159 revolution.

Positive position, counter-clockwise (CCW) motion.

Negative position, clockwise (CW) motion.

Distance $S = \theta \cdot r$. This different for each point on the object.

Displacement $\Delta\theta = \theta_2 - \theta_1$. Same for all points



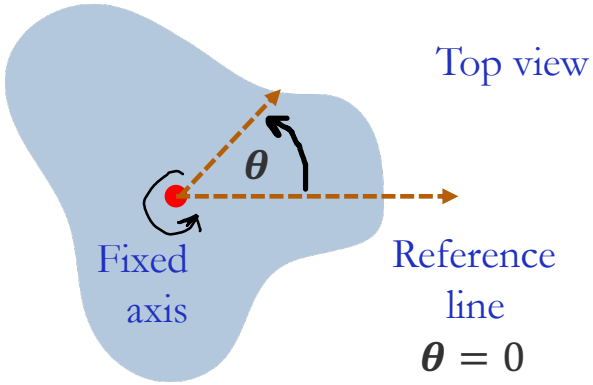
Angular Motion Variables

2) Average and instantaneous velocity

Average velocity $\omega_{ave} = \frac{\Delta\theta(t)}{\Delta t}$ (rad/s)

Instantaneous velocity (angular) $\omega = \frac{d\theta(t)}{dt}$ (rad/s)

Same for all parts of the object. Can be CW or CCW.



3) Average and instantaneous acceleration

Average acceleration $\alpha_{ave} = \frac{\Delta\omega}{\Delta t}$ (rad/s^2)

Instantaneous acceleration (angular) $\alpha = \frac{d\omega}{dt}$ (rad/s^2)

Same for all parts of the object.

4) Period

$$T = \frac{one\ cycle}{speed}$$

$$T = \frac{2\pi}{\omega} \quad (s)$$

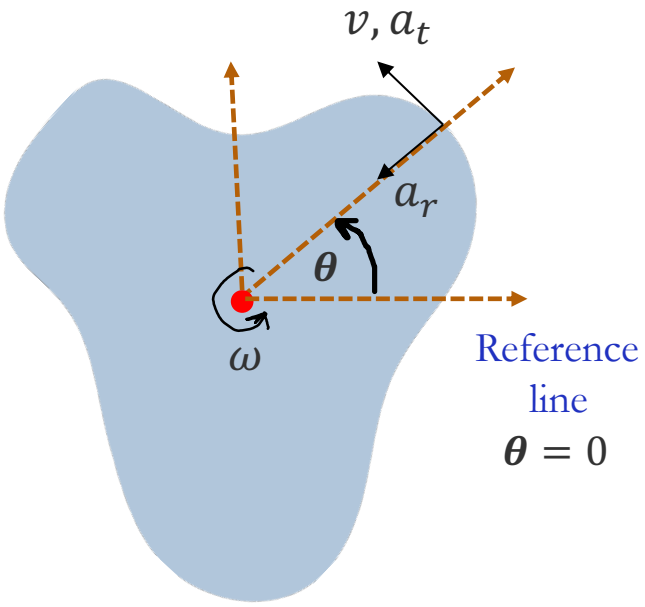
Equations of Angular Motion

For constant acceleration

$$\omega(t) = \omega_o + \alpha t$$

$$\theta(t) = \theta_o + \omega_o t + \frac{1}{2}\alpha t^2$$

$$\omega^2(t) = {\omega_o}^2 + 2\alpha \cdot [\theta(t) - \theta_o]$$



Angular-linear relations

$$S = \theta \cdot r$$

$$v = \omega \cdot r$$

$$a_t = \alpha \cdot r \qquad a_r = \omega^2 \cdot r$$

Examples

1) At t=0, the flywheel has an angular velocity of 4.7 rad/s and a constant angular acceleration of -0.25 rad/s^2. What is the maximum angle does the wheel go through before it stops? (reference is at t=0)

$$\omega^2(t) = {\omega_o}^2 + 2\alpha \cdot [\theta(t) - \theta_o]$$

$$\theta_{max} = \frac{\omega^2 - {\omega_o}^2}{2\alpha} = \frac{0^2 - 4.7^2}{-2(0.25)} = 44.18\, rad$$



How much time?

$$\omega(t) = \omega_o + \alpha t$$

$$0 = 4.7 + (-0.25)t$$

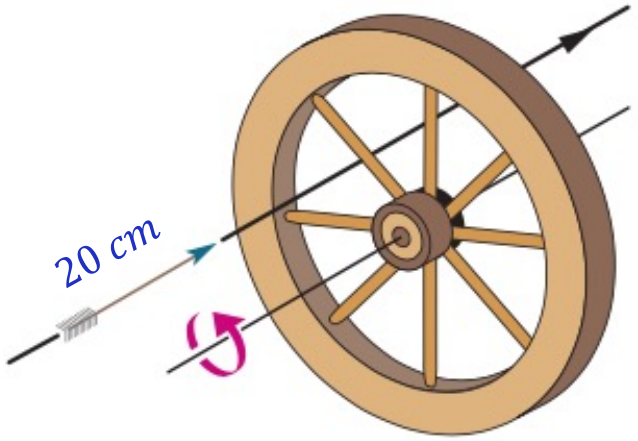
$$t = 18.8\, s$$

What happens after it stops?

Examples

2) The wheel has 8 equal slots. For the arrow to make a clean pass through, what should be its minimum speed?

$$\omega = 2.5\ \frac{rev}{s} \cdot \frac{2\pi\ rad}{1\ rev} = 5\pi\ \ rad/s$$



$r = 30\ cm$

$\omega = 2.5\ rev/s$

Time needed to pass = time between slots,

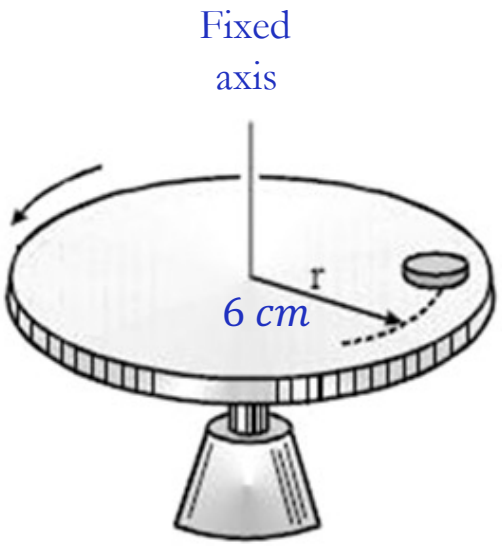
$$t = \frac{l_{arrow}}{v_{arrow}} = \frac{T}{8} = \frac{2\pi}{8\omega} = \frac{2\pi}{8(5\pi)}$$

$$v_{arrow} = (0.2)(20) = 4\ m/s$$

Where should the arrow be aimed?

Examples

3) The turning table is a rough surface and rotating at a constant speed. a) Find the radial acceleration of the coin. 2) the required static coefficient of friction between the coin and the table to prevent slipping.



$$\omega = 33\frac{1}{3}\frac{rev}{min} = 3.5\frac{\text{rad}}{\text{s}}$$

a) the radial acceleration $a_r = \omega^2 \cdot r$

$$a_r = (3.5)^2(6) = 73.3\ cm/s^2$$

b) static coefficient of friction

$$ma_r = \mu_s mg \qquad \mu_s = \frac{a_r}{9.81} = 0.075$$

c) Repeat (b) if the table has constant angular acceleration of 14 rad/s^2.

$$a_t = \alpha \cdot r$$

$$a_t = (14)(0.06) = 0.84\ m/s^2$$

$$a = \sqrt{{a_t}^2 + {a_r}^2} = \sqrt{0.84^2 + 0.733^2} = 1.115\ \ m/s^2$$

$$\mu_s = \frac{a}{9.81} = 0.114$$

Q & A

Send me an email if
you have any
questions.